

$$\Phi_n(x) = \prod_{\omega: 1 \text{ の原始 } n \text{ 乗根}} (x - \omega) \text{ とおく, } \Phi_n(x) = \frac{x^n - 1}{\prod_{d|n, d \neq n} \Phi_d(x)}$$

$$\Phi_1(x) = x - 1, \Phi_2(x) = x^2 + x + 1, \Phi_5(x) = x^4 + x^3 + x^2 + x + 1 \text{ はよくしるい,}$$

$$\Phi_{15}(x) = \frac{x^{15} - 1}{(x - 1)(x^2 + x + 1)(x^4 + x^3 + x^2 + x + 1)} = \frac{x^{10} + x^5 + 1}{x^2 + x + 1} = x^8 - x^7 + x^5 - x^4 + x^3 - x + 1.$$

$$\begin{array}{r} x^2 + x + 1 \overline{) x^{10} + x^9 + x^8} \\ \underline{x^{10} + x^9 + x^8} \\ -x^9 - x^8 \\ \underline{-x^9 - x^8 - x^7} \\ x^7 + x^6 + x^5 \\ \underline{x^7 + x^6 + x^5} \\ -x^6 \\ \underline{-x^6 - x^5 - x^4} \\ x^5 + x^4 \\ \underline{x^5 + x^4 + x^3} \\ -x^3 \\ \underline{-x^3 - x^2 - x} \\ x^2 + x + 1 \\ \underline{x^2 + x + 1} \\ 0 \end{array}$$

命題 p は素数であるとする. $\Phi_{15}(x) \bmod p$ は $\mathbb{F}_p$ 上既約でない. $\square$

例 $p = 2$ のとき, $\mathbb{F}_{16}^\times$ は 1 の原始 15 乗根 ω を含む. $F(a) = a^2$ ($a \in \mathbb{F}_{16}$) とおく. F -orbit of ω は

$$\omega \xrightarrow{F} \omega^2 \xrightarrow{F} \omega^4 \xrightarrow{F} \omega^8. \text{ ゆえに } f(x) = (x - \omega)(x - \omega^2)(x - \omega^4)(x - \omega^8) \in \mathbb{F}_2[x] \text{ 且 } f(x) \mid \Phi_{15}(x) \bmod 2 \quad \square$$

$n \in \mathbb{Z}_{>0}$, p は素数とする. 以下の (i), (ii) はすべて $\mathbb{F}_p$ 上の議論.

(i) $p \nmid n$ のとき. $n = p^k m$, $p \nmid m$, $k > 0$ と書ける. このとき,

$$\Phi_n(x) = \Phi_m(x)^{p^k - p^{k-1}} \quad (\text{特に } p^k > 2 \text{ ならば } \Phi_n(x) \text{ は既約ではない})$$

証明 $m=1$ のとき, $\Phi_p(x) = \frac{x^p - 1}{x - 1} = \frac{(x-1)^p}{x-1} = (x-1)^{p-1}$, $\Phi_{p^2}(x) = \frac{(x-1)^{p^2}}{(x-1)^p} = (x-1)^{p^2-p}$, ... でかつ

$$\Phi_{p^k}(x) = \frac{x^{p^k} - 1}{\Phi_1 \cdots \Phi_{p^{k-1}}} = \frac{(x-1)^{p^k}}{(x-1)^{1+(p-1)+\cdots+(p^{k-1}-p^{k-2})}} = \frac{(x-1)^{p^k}}{(x-1)^{p^{k-1}}} = (x-1)^{p^k - p^{k-1}}.$$

$0 < i \leq k$, $d \mid m$, $p^i d < n$ のとき, $\Phi_{p^i d}(x) = \Phi_d(x)^{p^i - p^{i-1}}$ が成立していると仮定する. このとき

$$\begin{aligned} \Phi_n(x) &= \frac{(x^m)^{p^k} - 1}{\prod_{\substack{1 \leq i \leq k, d \mid m \\ p^i d < n}} \Phi_{p^i d}} = \frac{(x^m - 1)^{p^k}}{\prod_{\substack{0 \leq i \leq k, d \mid m \\ p^i d < n}} \Phi_d^{p^i - p^{i-1}}} = \frac{(x^m - 1)^{p^k}}{\prod_{\substack{d \mid m \\ d < m}} \Phi_d^{p^k}} \times \Phi_m^{p^k - p^{k-1}} = \Phi_m(x)^{p^k - p^{k-1}}. \quad \text{g.e.d.} \\ &= \left(\frac{x^m - 1}{\prod_{d \mid m, d < m} \Phi_d} \right)^{p^k} = \Phi_m^{p^k} \end{aligned}$$

(ii) $p \mid n$ のとき

$x^n - 1$ は $\mathbb{F}_p$ 上重根を持たないので, $x^n - 1$ の根全体は位数 n の巡回群になる.

$$\text{その生成元を } \omega \text{ と書くと, } \Phi_n(x) = \prod_{\substack{\gcd(i, n) = 1 \\ 1 \leq i < n-1}} (x - \omega^i)$$

$\Phi_n(x)$ の最小分解体は $\mathbb{F}_p(\omega)$ になる. $F(a) = a^p$ とおく (Frobenius automorphism), このとき,

$\Phi_n(x)$ が $\mathbb{F}_p$ 上既約 $\Leftrightarrow \Phi_n(x)$ の根の全体は 1 つの F -orbit になる.

$\omega \xrightarrow{F} \omega^p \xrightarrow{F} \omega^{p^2} \xrightarrow{F} \cdots \xrightarrow{F} \omega^{p^{t-1}} \xrightarrow{F} \omega^{p^t} = \omega$ となる最小の正の整数を t と書く. このとき,

$$\Phi_n(x) \text{ が } \mathbb{F}_p \text{ 上既約} \Leftrightarrow t = \phi(n) \Leftrightarrow (\mathbb{Z}/n\mathbb{Z})^\times = \langle p \bmod n \rangle \Leftrightarrow (\mathbb{Z}/n\mathbb{Z})^\times \text{ は巡回群 } p \nmid n$$

以上の (i), (ii) を合わせると

$(\mathbb{Z}/n\mathbb{Z})^\times$ は巡回群でなく, $p \mid n$ または p が奇素数 $\Rightarrow \Phi_n(x)$ は $\mathbb{F}_p$ 上既約でない.

証明 $p \nmid n$ ならば (ii) より $\Phi_n(x)$ は $\mathbb{F}_p$ 上既約ではない.

$p \mid n$ かつ p が奇素数ならば (i) より $\Phi_n(x)$ は $\mathbb{F}_p$ 上既約ではない. g.e.d.

例 p', p'' は互いに異なる奇素数であるとし, $n = p'p''$ とおく.

このとき, $(\mathbb{Z}/n\mathbb{Z})^\times \cong C_{p'-1} \times C_{p''-1}$ は, $p'-1$ と $p''-1$ が共に偶数なので, 巡回群にならない.

したがって, $\Phi_n(x) = \Phi_{p'p''}(x)$ は $\mathbb{F}_p$ 上既約にならない.

特に $\Phi_{15}(x)$ は $\mathbb{F}_p$ 上既約にならない.

[18.5] Let n be a positive integer such that $(\mathbb{Z}/n)^\times$ is *not* cyclic. Show that the n^{th} cyclotomic polynomial $\Phi_n(x)$ factors properly in $\mathbb{F}_p[x]$ for any prime p not dividing n .

(See subsequent text for systematic treatment of the case that p divides n .) Let d be a positive integer such that $p^d - 1 = 0 \pmod n$. Since we know that $\mathbb{F}_{p^d}^\times$ is cyclic, $\Phi_n(x) = 0$ has a root in $\mathbb{F}_{p^d}$ when $p^d - 1 = 0 \pmod n$. For $\Phi_n(x)$ to be irreducible in $\mathbb{F}_p[x]$, it must be that $d = \varphi(n)$ (Euler's totient function φ) is the smallest exponent which achieves this. That is, $\Phi_n(x)$ will be irreducible in $\mathbb{F}_p[x]$ only if $p^{\varphi(n)} = 1 \pmod n$ but no smaller positive exponent achieves this effect. That is, $\Phi_n(x)$ is irreducible in $\mathbb{F}_p[x]$ only if p is of order $\varphi(n)$ in the group $(\mathbb{Z}/n)^\times$. We know that the order of this group is $\varphi(n)$, so any such p would be a generator for the group $(\mathbb{Z}/n)^\times$. That is, the group would be cyclic.

[18.6] Show that the 15^{th} cyclotomic polynomial $\Phi_{15}(x)$ is irreducible in $\mathbb{Q}[x]$, despite being reducible in $\mathbb{F}_p[x]$ for every prime p .

First, by Sun-Ze

$$(\mathbb{Z}/15)^\times \approx (\mathbb{Z}/3)^\times \times (\mathbb{Z}/5)^\times \approx \mathbb{Z}/2 \oplus \mathbb{Z}/4$$

This is not cyclic (there is no element of order 8, as the maximal order is 4). Thus, by the previous problem, there is no prime p such that $\Phi_{15}(x)$ is irreducible in $\mathbb{F}_p[x]$.

To prove that Φ_{15} is irreducible in $\mathbb{Q}[x]$, it suffices to show that the field extension $\mathbb{Q}(\zeta)$ of $\mathbb{Q}$ generated by any root ζ of $\Phi_{15}(x) = 0$ (in some algebraic closure of $\mathbb{Q}$, if one likes) is of degree equal to the degree of the polynomial Φ_{15} , namely $\varphi(15) = \varphi(3)\varphi(5) = (3-1)(5-1) = 8$. We already know that Φ_3 and Φ_5 are irreducible. And one notes that, given a primitive 15^{th} root of unity ζ , $\eta = \zeta^3$ is a primitive 5^{th} root of unity and $\omega = \zeta^5$ is a primitive third root of unity. And, given a primitive cube root of unity ω and a primitive 5^{th} root of unity η , $\zeta = \omega^2 \cdot \eta^{-3}$ is a primitive 15^{th} root of unity: in fact, if ω and η are produced from ζ , then this formula recovers ζ , since

$$2 \cdot 5 - 3 \cdot 3 = 1$$

Thus,

$$\mathbb{Q}(\zeta) = \mathbb{Q}(\omega)(\eta)$$

By the multiplicativity of degrees in towers of fields

$$[\mathbb{Q}(\zeta) : \mathbb{Q}] = [\mathbb{Q}(\zeta) : \mathbb{Q}(\omega)] \cdot [\mathbb{Q}(\omega) : \mathbb{Q}] = [\mathbb{Q}(\zeta) : \mathbb{Q}(\omega)] \cdot 2 = [\mathbb{Q}(\omega, \eta) : \mathbb{Q}(\omega)] \cdot 2$$

Thus, it would suffice to show that $[\mathbb{Q}(\omega, \eta) : \mathbb{Q}(\omega)] = 4$.

We should not forget that we have shown that $\mathbb{Z}[\omega]$ is Euclidean, hence a PID, hence a UFD. Thus, we are entitled to use Eisenstein's criterion and Gauss' lemma. Thus, it would suffice to prove irreducibility of $\Phi_5(x)$ in $\mathbb{Z}[\omega][x]$. As in the discussion of $\Phi_p(x)$ over $\mathbb{Z}$ with p prime, consider $f(x) = \Phi_5(x+1)$. All its coefficients are divisible by 5, and the constant coefficient is exactly 5 (in particular, not divisible by 5^2). We can apply Eisenstein's criterion and Gauss' lemma if we know, for example, that 5 is a prime in $\mathbb{Z}[\omega]$. (There are other ways to succeed, but this would be simplest.)

To prove that 5 is prime in $\mathbb{Z}[\omega]$, recall the *norm*

$$N(a + b\omega) = (a + b\omega)(a + b\bar{\omega}) = (a + b\omega)(a + b\omega^2) = a^2 - ab + b^2$$

already used in discussing the Euclidean-ness of $\mathbb{Z}[\omega]$. One proves that the norm takes non-negative integer values, is 0 only when evaluated at 0, is *multiplicative* in the sense that $N(\alpha\beta) = N(\alpha)N(\beta)$, and $N(\alpha) = 1$ if and only if α is a unit in $\mathbb{Z}[\omega]$. Thus, if 5 were to factor $5 = \alpha\beta$ in $\mathbb{Z}[\omega]$, then

$$25 = N(5) = N(\alpha) \cdot N(\beta)$$

For a proper factorization, meaning that neither α nor β is a unit, neither $N(\alpha)$ nor $N(\beta)$ can be 1. Thus, both must be 5. However, the equation

$$5 = N(a + b\omega) = a^2 - ab + b^2 = (a - \frac{b}{2})^2 + \frac{3}{4}b^2 = \frac{1}{4}((2a - b)^2 + 3b^2)$$

has no solution in integers a, b . Indeed, looking at this equation mod 5, since 3 is not a square mod 5 it must be that $b = 0 \pmod 5$. Then, further, $4a^2 = 0 \pmod 5$, so $a = 0 \pmod 5$. That is, 5 divides both a and b . But then 25 divides the norm $N(a + b\omega) = a^2 - ab + b^2$, so it cannot be 5.

Thus, in summary, 5 is prime in $\mathbb{Z}[\omega]$, so we can apply Eisenstein's criterion to $\Phi_5(x+1)$ to see that it is irreducible in $\mathbb{Z}[\omega][x]$. By Gauss' lemma, it is irreducible in $\mathbb{Q}(\omega)[x]$, so $[\mathbb{Q}(\omega, \eta) : \mathbb{Q}(\omega)] = \varphi(5) = 4$. And this proves that $[\mathbb{Q}(\zeta) : \mathbb{Q}] = 8$, so $\Phi_{15}(x)$ is irreducible over $\mathbb{Q}$.